Cereal 5 — Experiments

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Five steps were run: a monolithic baseline, threshold tuning on it, then a specialist redesign that was checked on one fold, adjusted, and run in full. The monolithic model with tuned thresholds remains the best result.

Experiment log

Step	What was done	Why	Macro-F1 (OOF)	Outcome
1	Monolithic ConvNeXt-Base, 8 sigmoids, weighted BCE, threshold 0.5	Baseline	0.432 (micro 0.492, Hamming 0.264)	Rare labels over-predicted: high recall, low precision
2	Per-label threshold search on step 1's OOF predictions	Fix the recall bias from weighted BCE	0.468 (micro 0.543, Hamming 0.204)	Best maize result
3	Specialist pipeline, fold 1 only	One shared head blurs 8 decision boundaries; SR has 39 positives against 2,316 negatives	0.334 (micro 0.356)	Already below monolithic
4	Compared S5's validation AUC by phase	Check whether fine-tuning helps S5	Frozen 0.8026; top-30 0.7830; full 0.7919	Fine-tuning removed from S5
5	Specialist pipeline, all 5 folds	Full evaluation	0.260 at 0.5; 0.283 tuned	Clearly worse than monolithic; open

What the specialist design assumed

- Binary specialists S1–S4 learn a sharp healthy-vs-disease boundary from clean images, free of competition with other labels.
- Grouping SR with other rare or vague labels in S5 gives it more signal than SR alone.
- Per-specialist thresholds can be tuned like the monolithic ones.

The result says at least one assumption fails. Two candidate causes, not yet tested: (a) S1–S4 never see images with several diseases, which are 38% of the data; (b) `train_specialist()` keeps the last phase's model even when an earlier phase was better (see Known Issues).

Rules for the next run

1. Fix the model-saving bug first, so you are not testing a broken pipeline.
2. Run the duplicate screen and use grouped folds.
3. Tune thresholds on an inner validation split, not the OOF predictions you report.
4. Report per-label F1, macro-F1, micro-F1, Hamming loss and AUC for both pipelines side by side.
5. Run fold 1 of any new design first and compare with the monolithic fold-1 score before spending 25 runs.